

Routledge
Taylor & Francis Group

Journal of Urban Technology

ISSN: 1063-0732 (Print) 1466-1853 (Online) Journal homepage: www.tandfonline.com/journals/cjut20

Mundane Urban Governance and AI Oversight: The Case of Vienna's Intelligent Pedestrian Traffic Lights

Pouya Sepehr

To cite this article: Pouya Sepehr (2025) Mundane Urban Governance and AI Oversight: The Case of Vienna's Intelligent Pedestrian Traffic Lights, Journal of Urban Technology, 32:3, 93-110, DOI: 10.1080/10630732.2024.2302280

To link to this article: https://doi.org/10.1080/10630732.2024.2302280

© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 23 Feb 2024.

Submit your article to this journal

Article views: 2123

View related articles

View Crossmark data

Citing articles: 8 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=cjut20

JOURNAL OF URBAN TECHNOLOGY
2025, VOL. 32, NO. 3, 93–110
https://doi.org/10.1080/10630732.2024.2302280

Routledge
Taylor & Francis Group

OPEN ACCESS Check for updates

Mundane Urban Governance and AI Oversight: The Case of Vienna's Intelligent Pedestrian Traffic Lights

Pouya Sepehr

Department of Science and Technology Studies, University of Vienna

ABSTRACT

This article examines the implications of integrating Artificial Intelligence (AI) into urban infrastructures, using Vienna's Intelligent Pedestrian Traffic Light (IPTL) as a focal point. As cities globally embrace AI, Urban AI (UAI) emerges, reshaping urban life. The central inquiry is: How does AI in urban governance alter urban spaces, and what new political dynamics arise? The IPTL, a UAI exemplar, uses advanced surveillance and algorithms to enhance pedestrian signaling, aiming to reduce jaywalking and optimize traffic. Through mixed-methods, combining theoretical and empirical approaches, the study delves into expert narratives, public perceptions, and the technology's design. It uncovers the transformative effects of such technologies on urban spaces and human–vehicle interactions. Findings highlight that while the IPTL promises urban governance advancements, it may overlook diverse urban needs, as seen in the neglect of the visually impaired. The research accentuates the imperative for a more encompassing, inclusive lens in AI-centric urban endeavors, underscoring the significance of recognizing and addressing the myriad rhythms and requirements of all urban residents. The article culminates in a call for a reconstructivist perspective on UAI, championing the democratization of urban realms and foregrounding pedestrian-centric visions in upcoming mobility ventures.

KEYWORDS

mundane urban governance; urban AI (UAI); AI-driven urban mobility; smart city; traffic light; democratizing urban technology

Introduction

The infusion of Artificial Intelligence (AI) into urban spaces is both deep-rooted and ever-evolving. Serving as a subtle yet influential force, AI orchestrates the multifaceted interactions between individuals and their urban environments (Ames, 2018). It is within these urban environments that AI innovations are birthed, honed, and integrated. Cugurullo (2020) introduces the concept of Urban Artificial Intelligences (UAI) to describe these AI systems tailored for the urban milieu. Such technologies are intrinsically linked to the city, drawing inspiration from it while simultaneously reshaping its

CONTACT Pouya Sepehr pouya.sepehr@univie.ac.at Department of Science and Technology Studies, University of Vienna. Universitätsstraße 7/Stiege II/6. Stock 1010 Vienna, Austria

© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

essence and operations. This symbiotic evolution underscores a significant shift in urban paradigms, marking a journey from mere automation to all-encompassing autonomy.

As cities become more intertwined with advanced AI infrastructures, understanding the nuances of this urban–technological convergence becomes paramount. While theoretical perspectives provide a foundation, there is a compelling call for empirical studies to delve into UAI's tangible effects on urban life. Such research endeavors can illuminate the transformative influence of AI on urban inhabitants, administrative structures, and the overarching urban socioeconomic fabric. Through rigorous empirical scrutiny, researchers can navigate the intricacies of UAI, addressing its challenges and harnessing its potential. This study embarks on such a journey, focusing on a specific AI-driven urban technological intervention in Vienna, asking: how does the integration of AI into mundane urban governance reshape the everyday life of urban spaces, and what political configurations and power relations emerge from these technological interventions? Furthermore, how can we ensure sensitivity and inclusivity in recognizing and addressing the new capabilities and power dynamics introduced by such technologies?

Embedded within the fabric of our urban environments, many AI systems seamlessly integrate with tangible assets such as computers, vehicles, and infrastructure, becoming indispensable to their functionality (Cugurullo, 2021). A prime illustration of this integration is Vienna's Intelligent Pedestrian Traffic Light (IPTL) initiative. This initiative harnesses the power of sophisticated surveillance and algorithms to automate pedestrian signaling, aiming to deter jaywalking and optimize traffic flow. Within the IPTL framework, the term "intelligence" encapsulates the synergy of advanced surveillance and algorithmic traffic management. This innovative system possesses the capability to detect and recognize entities, anticipate their trajectories, and evaluate behavioral patterns at crossroads. Significantly, the IPTL is deployed at intersections equipped with "sans button" pedestrian lights, where pedestrians traditionally had the onus of manually activating the signal. In this scenario, the AI-driven system supersedes the manual activation, streamlining the pedestrian crossing experience. While the underlying algorithms of the IPTL might appear rudimentary at first glance, this seemingly commonplace technology holds the potential to profoundly reshape citizens' interactions with their urban surroundings. This article seeks to unravel the complex nexus between technology, society, and urban landscapes, highlighting how seemingly ordinary technological integrations, when fused with AI, can wield a transformative influence far beyond their apparent simplicity.

Building on this, it is essential to recognize that the pervasive infusion of AI into our daily lives, steered by algorithmic governance, challenges traditional urban paradigms and paves the way for innovative space reconfigurations. The conventional "city-as-computer" metaphor, which views urban complexities as mere programmable entities, is now under scrutiny (Mattern, 2021: 62). Furthermore, urban technological innovations often reflect the visions of major corporations rooted in tech-solutionism. These innovations, with their overt and covert intentions, can exert amplified control over societal structures (Sadowski, 2020; Sepehr and Felt, 2023). Vienna's IPTL exemplifies these technological interventions, emphasizing the transformative role of AI in urban mobility dynamics. This evolution compels us to reevaluate the nature of urban spaces birthed from such technological progress (Tironi and Valderrama, 2018).

In light of this, the research aims to dissect the IPTL's emergence in public space, with an emphasis on the nuances of visibility and invisibility throughout its inception and deployment. Through a blend of theoretical frameworks and empirical exploration, this research elucidates the profound impacts of AI on urban dynamics, governance paradigms, and the overarching socio-cultural milieu. Adopting a mixed-methods strategy, the study scrutinizes expert discourses, their perception of affected publics, and the intricate design and rollout of the technology employed to solve a mundane urban problem. It reveals the profound ripple effects of such seemingly innocuous technologies on urban landscapes, human–vehicle interplay, and the city's intricate socio-technical weave.

Key findings spotlight that while initiatives like the IPTL herald improved urban governance and mobility, they can inadvertently sideline the multifarious needs of urban dwellers, exemplified by the oversight of the visually impaired in the IPTL's blueprint. The research accentuates the imperative for a more encompassing, inclusive lens in AI-centric urban endeavors, underscoring the significance of recognizing and addressing the myriad rhythms and requirements of all urban residents. The article culminates in a call for a reconstructivist perspective on UAI, championing the democratization of urban realms and foregrounding pedestrian-centric visions in upcoming mobility ventures.

Research Approach

This research embarks on an in-depth exploration of the IPTL, delving into its genesis, design, and implementation. Central to this investigation is the dynamic interrelation between its technical components, societal participants, and the governance of urban mobility. By mapping out the constellation of actors involved—ranging from designers and policymakers to everyday pedestrians and other key stakeholders—the study aims to unravel the complex web of relationships that gave rise to the IPTL.

Approaching from a Science and Technology Studies (STS) perspective, two facets stand out as crucial in understanding the societal integration of technology. First, technologies are not mere tools or instruments. Instead, they are intricate constructs that encapsulate and reflect the aspirations, ideologies, and values of their creators (Akrich, 1992; Latour, 1992). These constructs, far from being neutral, embody specific attributes and design choices that shape their role, influence, and interactions within distinct social contexts. They become entwined with societal norms, practices, and structures, often influencing and being influenced by them.

Second, technologies are dynamic entities, not fixed or immutable constructs. They emerge and evolve from complex material-semiotic interconnections, which are networks of relationships between people, objects, ideas, and practices (Law, 2004; Latour, 2005). These interconnections, formed by a diverse array of actors—from individual enthusiasts and experts to institutional bodies and policymakers—collectively define the technology's trajectory, adoption, and adaptation in society. Within these networks, technologies are continually negotiated, contested, and redefined, leading to shifts in their societal roles and implications. Recognizing this dynamism is essential for understanding the broader impacts and potential futures of technologies like the IPTL in urban settings.

Data collection for this study employs a multifaceted approach, incorporating in-depth interviews, direct observations, and comprehensive document analysis. The research commenced with an exhaustive review of publicly accessible materials, including social media posts and newspaper articles, spanning from 2019 to 2021. This review focused on identifying the key motivations and driving forces that led to the IPTL's implementation. To achieve this, the study consulted a variety of sources, such as urban policy documents, scholarly articles, and established traffic management guidelines. Noteworthy references include the UN Convention on Road Traffic, Geneva, 1949; seminal academic works on the evolution of traffic lights; the Vienna Urban Mobility Plan STEP-2025, published in December 2014; the Viennese Traffic Light Signaling System (Verkehrslichtsignalanlagen RVS 05.04.31, initially released in 1960 and updated in 1994); and the program "Mobility of the Future" (Mobilität der Zukunft [MdZ])—RTI Agenda and Roadmap for Personal Mobility, released in Vienna in August 2018.

To complement the document analysis, public relations materials from the IPTL project and four academic papers from the computational design team were scrutinized. These sources provide valuable insights into the technical aspects of the IPTL, its compliance with existing regulations, and the envisioned role of AI in shaping Vienna's future mobility landscape. Media coverage from leading Austrian publications like *Kronen Zeitung*, *Der Standard*, and *Kurier*, along with radio segments from ORF.at, further enrich our understanding of the IPTL's local implementation. Social media platforms, particularly *Twitter*, were also scoured to capture public sentiment and firsthand experiences related to the IPTL.

To augment the document-based research, the study incorporates direct observations of the IPTL prototypes deployed at five distinct locations across Vienna. Over a span of 10 hours, field visits were conducted between March and September 2021. These visits focused on understanding the temporal dynamics at intersections where the IPTL was operational. Observations were made to capture the diverse rhythms of pedestrian movement, identifying various types of users and modes of mobility. The objective was to gain a nuanced understanding of how pedestrians interact with the IPTL in real-time, shedding light on behavioral patterns and temporal dynamics.

For a more in-depth technical perspective, the study conducted comprehensive interviews with six subject-matter experts who played pivotal roles in the IPTL project. These interviewees, selected because of their specific contributions to the IPTL's design, included faculty from the Technical University of Graz and professionals from Vienna's Traffic Management Department. These dialogues aimed to capture expert viewpoints, decision-making frameworks, and the socio-technical factors that influenced the IPTL's design. The insights gleaned from these conversations provide a rich contextual backdrop for interpreting the IPTL's impact on pedestrian behavior and temporal rhythms.

The subsequent sections of this paper will situate the analysis within a historical context, tracing the evolving roles of traffic lights in urban spaces and their transformation, which has often favored vehicular movement over pedestrian mobility. Building on this historical groundwork, the research will delve into the contemporary functions of traffic lights as temporal regulators. This will culminate in a detailed examination of the IPTL's current role, significance, and broader implications in the urban landscape of Vienna.

Traffic Lights and the Socio-Technical Reconfiguration of Streets and Urban Spaces

As ubiquitous fixtures of modern urban landscapes, traffic lights have become emblematic of organized society, achieving a level of global standardization that few inventions can claim (McShane, 1999). They serve as beacons, exemplifying the potential of a unified global law and order system (Zeno-Zencovich, 2016). The rise and widespread acceptance of traffic lights weave an intriguing narrative of urban transformation: streets evolving from pedestrian-dominated spaces to vehicular thoroughfares. Peter Norton's *Fighting Traffic* (2011) offers a vivid account of this metamorphosis in American streets during the early twentieth century. With the automobile boom, the 1920s witnessed a disturbing surge in fatal accidents. This escalating mortality rate sounded alarm bells, sparking media debates and prompting calls for authoritative interventions. The ensuing public discourse grappled with fundamental questions about the very essence of streets and their intended users (Norton, 2007: 331).

Various societal factions, each armed with their unique visions and values, entered the fray with proposed solutions. The Motordom lobby group, representing the American Automobile Association, emerged as a formidable voice. They painted the automobile as a symbol of freedom and a beacon of American progress (Norton, 2014). Their campaigns championed the vision of streets as streamlined arteries for vehicles, advocating for the marginalization of pedestrians. In this vehicular utopia, pedestrian traffic lights became instrumental, designed to shepherd pedestrians and keep roads clear. The era also birthed the term "jaywalkers," a pejorative label for those flouting traffic norms, essentially branding them as urban novices which literally refers to "someone who did not know how to walk in a city"[1] (Norton, 2014: 25).

Traffic lights, while conceived as instruments of safety and order, inadvertently sculpted new societal norms. With their advent, law enforcement found a new target: the burgeoning tribe of "misbehaving" jaywalkers, now liable to fines. This binary of compliance and defiance, crystallized by traffic lights, delineated the contours of civic behavior. Beyond mere regulation, traffic lights became pivotal cogs in the larger machinery of urban transformation, reshaping streets for the age of automobility (Sheller and Urry, 2000; Braun and Randell, 2022a). They didn't just regulate; they redefined urban spaces. Symbols like traffic lights and the jaywalker[2] not only facilitated urban transitions but also rendered them palpable. As such, they warrant democratic scrutiny, for they shape not just streets, but the very ethos of urban life.

The transformation of streets into vehicular domains is but one manifestation of broader spatial and environmental evolutions. Mumford (1964) had presciently warned of the potential pitfalls of unchecked highway expansion and rampant automobile reliance. He envisioned urban centers choked by an unyielding tide of vehicles and crumbling infrastructure. Today, the automobile's environmental footprint is undeniable, with significant contributions to CO_2 emissions and the monopolization of vast urban expanses. Streets, which can constitute up to a third of a city's built environment, are overwhelmingly designed with vehicles[3] in mind. This car-centric approach has levied a heavy toll: streets have metamorphosed into arenas of congestion, accidents, trauma, and even political strife (Braun and Randell, 2022b).

Yet, there's a burgeoning movement advocating for more humane, livable urban spaces.[4] Activists, urban planners, and visionaries are rallying behind a paradigm shift in the transportation ethos. Ideas like car-free zones, 15-minute cities, and open streets are no longer utopian dreams but actionable blueprints. These contemporary urban narratives, in their essence, champion the reclamation of streets from the clutches of vehicles, emphasizing the creation and expansion of public spaces (refer to Sheller [2018] for an in-depth discourse on twenty-first-century mobility justice). However, amidst these transformative discussions, the role of traffic lights, especially in the context of emerging traffic management technologies, has been somewhat eclipsed. This article endeavors to bridge this gap, casting a spotlight on traffic lights as crucial temporal orchestrators of urban dynamics.

Traffic Lights and the Temporal Ordering of Space

The intricate dance between time and space has historically been overshadowed in urban studies. Monstadt (2022) argues that adopting a temporal lens, one attuned to urban rhythms, can sharpen our understanding of the transformative power of infrastructures. These infrastructures, he suggests, are not just static entities; they carry the weight of historical legacies while also acting as conduits for present aspirations and visions of the future. This temporal continuum often becomes a battleground, with established infrastructures clashing against the allure of nascent technologies. The IPTL stands as a testament to this tension, promising AI-driven efficiencies while still tethered to conventional urban traffic management systems.

At their core, traffic lights are instruments of prioritization (Coletta et al., 2020). Their design inherently favors certain modes of transport, with time allocations reflecting these preferences. As orchestrators of urban rhythm, traffic lights choreograph the ebb and flow of humans and vehicles within cityscapes. The IPTL, in this context, emerges as a focal point of divergent interests, each vying for temporal precedence and catering to varied mobility paradigms. Coletta and Kitchin (2017) delve into the transformative potential of computational algorithmic management on urban cadences. Drawing from Miyazaki's (2013) notion of "Algorhythms," they coin the term "algorhythmic governance" to dissect how smart city initiatives calibrate and regulate the multifarious temporal beats of urban life. Their work challenges conventional notions of rhythm, urging a reevaluation of measures, linearity, and cyclical patterns as dynamic constructs (Coletta and Kitchin, 2017: 4). The myriad interpretations and applications of these rhythms, given diverse urban contexts, underscore the need to probe deeper into the power dynamics and controls that temporalities exert.

The politics of mobility invariably leave their imprints on urban spaces (Cresswell, 2006; Sheller, 2017). As Jensen (2011) posits, mobility isn't monolithic; its myriad forms each encapsulate a unique "power dynamic," emphasizing specific modes of movement within urban confines. Jensen's work calls for a holistic appreciation of mobility's power dynamics, spotlighting the practices, rationalities, and emotions that underpin it, all deeply embedded within the socio-political tapestry of urban settings. Concurrently, the diverse interpretations of ubiquitous governance tools, like traffic lights, across different cities present a fascinating study in contrasts (Woolgar and Neyland, 2013).

Take, for instance, the behavior of pedestrians in Vienna, who typically respect red signals, even in the absence of oncoming traffic. Woolgar and Neyland (2013) posit that such governance practices, tied to everyday artifacts like the IPTL, mirror the politics and entrenched norms of collective behavior. They further argue that the socio-technical dimensions of daily life are intrinsically linked to rule-setting, monitoring, and adherence (6). However, the rich tapestry of STS traditions beckons us to challenge these established norms, urging us to envision alternative paradigms for technological phenomena. As Woolgar and Neyland succinctly put it, "it could be otherwise" (7). This perspective fosters a healthy analytical skepticism, nudging us to probe deeper into the political, scientific, and material choices that shape the genesis of new technological artifacts.

With this foundational understanding, the ensuing section will delve into the IPTL's role in orchestrating mobility within urban spaces, spotlighting the novel dynamics and ramifications introduced by AI in automating routine urban tasks.

IPTL Unveiled: From Conceptualization to Urban Trials

Justifications to Initiate the IPTL

Vienna operates approximately 1,300 pedestrian traffic lights, of which 200 employ a "sans button" system. These traffic lights require pedestrians to press a button to activate the signal, allowing them to cross. Locally, these are referred to as "Bettlerampel," translating to "beggars' traffic light." They are typically found on busy roads with infrequent pedestrian traffic, activating only upon request. A traffic engineer from the City of Vienna highlighted a challenge with the sans button system, noting, "they often lead to misunderstandings and confusion since pedestrians anticipate the signal to change automatically." To address this, the Traffic Light Department of the City Administration (MA 33) collaborated with the Computer Vision and Surveillance Lab of the Technical University of Graz (TUG) to develop a system that detects pedestrians intending to cross, thereby changing the signal without manual activation (Ertler et al., 2018).

Elaborating on the issue, an expert from the research group shared, "Initially, pedestrians might be unaware that they need to activate the traffic light using the sans button system, which can cause delays." The confusion is exacerbated as the button box bears a resemblance to the acoustic signal systems designed for the visually impaired. The expert added, "Even if a pedestrian activates the button, they might cross immediately if the road appears clear, effectively jaywalking." This behavior has a cascading effect: "After the individual crosses, the light turns red, halting vehicular traffic even though no pedestrian is present." The primary concern is the unnecessary delay for both vehicles and pedestrians unfamiliar with the pushbutton system. Summarizing the objective, the TUG expert stated, "to get rid of the pushbutton box [sans button] and replace it with a more reliable and automated system."

Design Protocols

To address the sans button system's challenges, MA33 invited the TUG team to propose a solution using intelligent surveillance technology. Initially, the research team conducted a feasibility study. As one interviewee explained, the study's purpose was to establish a

shared understanding with the client, document the project's objectives, estimate the budget, and outline the implementation steps. In essence, the feasibility study reviews current state-of-the-art techniques to present various technical options aligned with the project's aim. This study acts as an *inscription manual* (Akrich, 1992) that "define [s] a framework of action together with the actors and the space in which they are supposed to act in" (Akrich, 1992, 208). Thus, it delineates—or in Akrich terminology it *describes*—the technology's purpose, its users, and the ideals, morals, and values it embodies. Unfortunately, social scientists who want to get hold of the feasibility study will face fierce rejection, so this study cannot say much about it.

Nevertheless, it was possible to comprehend how certain socio-political presumptions influenced the design protocol and technical development. For instance, privacy concerns were high, given the use of smart camera surveillance technology, which complicated the project. To address this as a requirement, the team adopted a "privacy by design" approach, ensuring all data remained within the camera to prevent potential image leaks. This design choice posed technical challenges, as finding a reasonably priced processor capable of analyzing data from a high-resolution camera proved difficult. The team had to balance camera resolution quality with the processor's capabilities. Media coverage of the innovation consistently highlighted the project's success in preserving privacy, indicating that public sensitivities influenced the technology's development.

The TUG research team, during the research phase, introduced a more advanced prediction system that offered features beyond the initial proposal. Utilizing global movement models and recorded data, they developed learning algorithms that detect a pedestrian's intention to cross even before reaching the traffic light. As one TUG expert explained, the camera can identify everyone "within an eight by five meters field of view in seconds!" The rapid processing mechanism "takes one second for an initial estimate, which becomes reliable after just two seconds, resulting in a total prediction duration of less than four seconds." This speed surpasses manual activation leading to a more efficient traffic flow, which is a desired outcome for the traffic management purpose.

The TUG researcher further detailed three primary functions of the new system, extending beyond the project's initial objectives: (1) If pedestrians are in the designated waiting zone, the signal automatically turns green; (2) If a pedestrian departs before the signal turns green, the signal change halts, allowing uninterrupted vehicular flow; and (3) For larger groups, such as schoolchildren at peak times, the green phase extends automatically. These functions emerged from lab experiments, which not only sought the best technical combination within the project's scope but also unveiled new technical capabilities. Thus, the lab experiments illuminated new possibilities and capacities.

The Street Trial of IPTL

Upon finalizing the best configuration, the next step involved creating a prototype and initiating a trial phase in public spaces. Drawing from Marres' case study on intelligent vehicle testing in the UK (2020a, 2020b), streets serve as public trial spaces for introducing societal innovations. She identifies this as a "double-edged operation" effect (Marres, 2020b: 113). On one side, it is about engaging with citizens, inviting them to interact with

the new artifact, which in turn helps designers gather feedback. This feedback encompasses the public's use, opinions, needs, desires, and even potential rejections of the artifact. On the flip side, the trial also serves to normalize the new artifact in the public's daily life, making it a familiar entity. Marres (2020a) champions a "re-constructive approach" in design protocols, emphasizing the importance of how social aspects are rendered visible or become obscured during tests (Marres, 2020a: 540). The IPTL's street trial phase mirrors Marres' findings. While the trial was designed to have a double operation effect, it seemed to miss the mark on capturing the full social dynamics in its design protocol.

The IPTL trial was conducted at five different crossings throughout the city.[5] These locations were strategically chosen to represent a variety of crosswalk situations. For instance, one was near a busy intersection adjacent to a metro station, while another was situated in front of a bustling shopping mall. The trial phase's primary goal was to usher the IPTL into society, aiming to replace the older technology with a more advanced AI-automated system. However, soon after the trial's initiation, two significant challenges emerged.

First, the city of Vienna received complaints from the Association for Early Intervention for the Blind, Visually Impaired, and Multiple Visually Impaired People. Given that all pedestrian traffic lights in Vienna are equipped with a sonic guide system that closely resembles the sans button box, this presented a challenge. The sonic guide system had to be incorporated, which directly conflicted with the project's aim of eliminating the box. Secondly, despite the system's rapid ability to detect a pedestrian's intention to cross, a safety margin was still mandated before the traffic signal could change. This requirement rendered the system's impressive 4-second detection time moot in practical scenarios.

The trial phase illuminated a disparity between the technology's theoretical promises and its real-world delivery. The technology, which promised efficiency and reduced confusion, fell short due to stringent safety regulations and a narrow understanding of its user base. Notably, the needs of visually impaired pedestrians were overlooked, effectively rendering them an "invisible" user in the design process. This oversight might have been addressed earlier if the feasibility study had been open to public–expert commentary. Instead, the experts appeared to have had a homogenized view of users.

In response to these challenges, two new developmental paths were proposed, leading to a subsequent project. First, engineers sought to integrate higher-resolution cameras and faster processors to enhance the system's ability to detect individuals with disabilities. Second, to amplify the system's impact, it was proposed that the entire street be integrated into the system. This integration would facilitate broader data collection, optimizing traffic flow across multiple intersections. For the engineers, improvement was synonymous with enhancing the system's detection capabilities. However, this was strictly from a technological perspective.

Throughout this process, a myriad of technical and social (in)visibilities came to the fore. A combination of socio-technical elements steered a series of decisions and adjustments in the design protocol and prototyping. The initial design protocol's oversight of certain user needs prompted a reevaluation of both design and implementation. This oversight is reminiscent of a classic social deficit model, as described by Marres (2020a). To address this, the study proposes a reconstructive approach, reimagining the otherwise sans button system.

The "Mission Impossible" of Coordinating Disparate Rhythms of Mobility

The complexities surrounding the IPTL's deployment necessitated a deeper exploration into pedestrian behavior vis-à-vis traffic management. This prompted empirical observations throughout the city, with a particular emphasis on the sans button crosswalks. The objective was to understand the multifaceted interactions pedestrians have with these crosswalks, grounding the broader theoretical discourse on traffic management in tangible experiences. During these observations, pedestrians at sans button crosswalks exhibited four primary behaviors:

- *Informed Locals*: These individuals, familiar with the sans button system, promptly press the button upon reaching the crosswalk and patiently await the green signal.
- *Rhythm Readers*: While acquainted with the sans button, this group chooses not to engage with it. Instead, they rely on their intuition of the traffic rhythm, crossing when they perceive it is safe.
- *Car-Interrupters*: These pedestrians press the button but don't necessarily wait for the green signal. They venture across when they see a gap in traffic, potentially causing delays for the cars. Some in this group might intentionally disrupt car flow, subtly protesting the perceived vehicular dominance on roads. The introduction of IPTL could curtail this form of silent protest.
- *Uninitiated Crossers*: Often tourists or newcomers, these individuals are oblivious to the sans button system. They either depend on other pedestrians to initiate the crossing signal or seize a natural traffic lull to cross.

These behaviors highlight the pivotal role of local knowledge. The first three groups, acquainted with the system, modulate their actions based on situational cues. The fourth group, however, often misses the sans button, possibly due to its unobtrusive design. Further observations indicated that pedestrian signal timings are frequently influenced by preceding traffic lights. Designed to minimize traffic disruption, this often results in a 40–60 second lag between the last car passing and the pedestrian light turning green. This delay, though seemingly trivial, can be a source of frustration for pedestrians.

Upon consultation with an MA33 traffic engineer, multiple factors contributing to this delay were identified. Regulatory safety margins during signal transitions can account for up to 15 seconds. Moreover, traffic flow calculations, based on average car speeds, can inadvertently extend this delay. In Vienna, some drivers have discerned that by marginally exceeding speed limits, they can bypass successive red lights, effectively becoming "*Jaydrivers*."

An MA33 traffic control manager described traffic management's intricacies as akin to a "mission impossible." The crux of the challenge is harmonizing the needs of diverse road users—cyclists, motorists, pedestrians, and public transport—within the confines of road space and signal timings. He articulated the dilemma: "We must cater to varied stakeholders: cyclists, motorists, pedestrians, public transport. Yet, we're constrained by limited road space and green light duration. Merging these constraints while satisfying everyone? It's our mission impossible." The stringent traffic light signal regulation (VLSA-RVS 05.04.31) in Vienna, which prescribes a 90–100 second signaling cycle for all traffic lights, exacerbates this challenge. This rigid cycle restricts

adaptability, prompting drivers to adjust speeds to align with the green phase and causing undue waiting times for pedestrians.

Here, the mission impossible of traffic control and the IPTL's deployment complexities, coupled with the intricate dance of pedestrian behaviors, prompted a deeper dive into Vienna's overarching traffic management philosophy. A candid conversation with a traffic engineer from TUW unveiled the formidable challenges of revamping the city's traffic management blueprint. The engineer spotlighted the dual hurdles of financial and systemic constraints. He explained that revamping the signaling program in Vienna is no small feat. "It's not just about tweaking a few settings; it necessitates an overhaul of the entire system, and that comes with a hefty price tag," he remarked. Delving into the intricacies of weaving human behavior into traffic models, he added, "Data collection is one thing, but distilling that into a simplified, rational model is another. Humans, by nature, don't always tread the path of logic."

The above quote points to the inherent challenges of integrating human behaviors into algorithmic traffic management systems. The traffic engineer's insights highlight the limitations of current algorithmic models in capturing the unpredictable and often illogical nature of human behaviors. While data collection can provide a snapshot of pedestrian movements, translating this data into a coherent and simplified model is fraught with complexities. This is primarily because human behaviors, unlike algorithmic processes, are not always linear or logical. The statement, "Humans, by nature, don't always tread the path of logic," serves as a poignant reminder of the unpredictability of human actions. It suggests that while algorithms operate within predefined parameters and logic, humans often act based on emotions, instincts, or other factors that may not always align with algorithmic predictions. This divergence between human behavior and algorithmic logic poses significant challenges for traffic management systems like the IPTL.

Furthermore, the mention of the "mission impossible of traffic control" and the need for a complete "overhaul of the entire system" emphasizes the magnitude of the challenge. It is not just about integrating new technology or tweaking existing systems; it is about reimagining the entire framework to accommodate the unpredictable nature of human behaviors. In contrast, the aim of the study is to call for a more open-ended approach to traffic management that acknowledges the complexities of human behaviors. While algorithmic solutions offer precision and efficiency, they must be designed with the understanding that human behaviors are multifaceted and cannot always be distilled into simplified models.

Historically, traffic models have been somewhat myopic, glossing over the rich tapestry of human adaptability and behavior. As the expert from TUW puts it: "Models, by design, truncate reality to make it manageable. They might omit certain variables for predictability or sheer difficulty in quantification. When the network evolves, the model perceives it as a brand-new entity, treating it as a static reality," the engineer elucidated. This modus operandi is deeply influenced by data collection methodologies and foundational understandings of transportation modes. Such a lens risks cementing a singular, perhaps skewed, version of reality, proposing solutions to problems that might be misinterpreted or misperceived.

Yet, the allure of technology's potential often doesn't metamorphose into a visionary redefinition of future traffic dynamics. This disconnect can be traced back to the

conventional role traffic lights have anchored in traffic orchestration. Frey et al. (2011) cast a critical eye on Vienna's traffic infrastructure, deeming it both inefficient and skewed in its priorities. The system's car-centric lens fails to embrace the fluid, evolving dance of traffic that morphs with time. Frey and his team posit that an overly software-driven approach in traffic management overlooks a fundamental truth: "Humans possess a remarkable adaptability to the ever-shifting tapestry of streets, an adaptability often underrepresented in software algorithms" (Frey et al., 2011: 73). This human adaptability manifests in behaviors like jaywalking, a testament to pedestrians' nuanced reading of traffic rhythms and street ambiance.

The Dance of Streets: Navigating a Traffic Light-less Vienna

In the heart of Vienna, as I embarked on this research journey, an unexpected ethnographic opportunity arose. A massive reconstruction project, paving the way for a new metro line, had taken over the vicinity of the department. This construction led to a cascade of changes: road blockages, tramline redirections, and the reshuffling of streets, sidewalks, and bicycle lanes. The process was tumultuous, with configurations being altered multiple times due to unforeseen challenges. One of the most intriguing outcomes of this upheaval was the temporary deactivation of traffic lights at two key intersections. This brief hiatus from automated traffic regulation transported the streets of contemporary Vienna back to a pre-traffic light era. As an ethnographer, I couldn't resist the allure of this natural experiment.

The intersection under my lens was where Auerspergstraße/Landesgerichtsstraße meet Staiongasse/Josefstädter Straße, adjacent to the Rathaus/Universität metro station. This intersection serves as a vibrant nexus, bustling with commuters en route to the municipality, university, and city center. Vienna's enchanting allure is especially evident along Josefstädter Straße, a historic avenue adorned with charming cafes and boutique stores. While Tram Line 2 meanders through, its presence discourages car traffic. Ideally, this would transform the area into a pedestrian paradise. However, the street's narrow confines combined with its popularity creates challenging congestion, making it a tricky terrain for both pedestrians and cyclists.

In this dynamic setting, the traffic light plays a crucial role in orchestrating the dance of different modes of mobility and pedestrians. As a regular commuter, I often found myself at odds with the traffic light's timing, leading me to seek alternative routes to avoid the wait. But with the lights out, a police officer stepped in, reminiscent of a bygone era. The officer, positioned at the heart of the intersection, engaged in a delicate ballet with the vehicles, the tram, and the pedestrians. Their gestures, a choreographed signaling with hands rising and falling, seemed intuitive, and even those unfamiliar with this old-world method seemed to grasp its rhythm.

The atmosphere at the intersection underwent a palpable shift. Every individual, whether on foot, bike, or in a car, became more attuned to their surroundings. The usual automated routine was disrupted, replaced by human interactions and shared acknowledgments. The police officer's presence, while ensuring order, also brought about a sense of community, reminiscent of a tourist's wonder when navigating an unfamiliar city. However, this human-centric approach wasn't without its challenges. The police officers, despite rotating shifts, had to be constantly vigilant. Their authority

was also more pronounced, as evidenced when a jaywalker, possibly under the influence, brazenly ignored the officer's signals and faced immediate reprimand.

This observation brought to mind Clay McShane's (1999) study on the origins of traffic control signals. McShane highlighted an often-overlooked aspect of traffic lights in America: their role in reducing police presence on streets, which had implications for black communities frequently subjected to police harassment. In this sense, traffic lights reduced racism and police harassment on the street bringing the effected communities a more engaging experience of being in the urban space. Intriguingly, the automated traffic light's inventor, Garrett Morgan, was black, adding another layer to the complex interplay between socio-technical artifacts and societal structures that leads to certain forms of invention and not others.

As we delve deeper into this article, we circle back to the central theme of STS politics and "otherwiseness." This vignette serves as a poignant reminder of the multifaceted nature of traffic management. It beckons us to reimagine and contemplate how a reconstructivist approach, especially in the context of IPTL, might manifest, emphasizing the need for a more inclusive and adaptive system that truly understands its users.

Reimagining IPTL and the Dynamics of Urban Mobility Otherwise

The urban landscape stands as a testament to human adaptability and the inherent flexibility of our movements. Leth et al. (2014) advocate for the decriminalization of jaywalking, extending this leniency not just to pedestrians but also to cyclists. Their research highlights how traffic regulations and modeling, as exemplified by systems like Vienna's traffic lights, have historically marginalized non-motorists. The call to action is clear: future traffic modeling must transcend the narrow confines of automobility (Sheller, 2017; Braun and Randell, 2022a). This calls for a paradigm shift from car-centric mobility to a more inclusive model that prioritizes human rhythms. It is about reimagining spaces where pedestrians are central, not marginalized as jaywalkers due to outdated regulations. In this context, there's an imperative to rethink and redesign the role of traffic lights in urban mobility.

One visionary who dared to think differently about traffic management was Dutch traffic engineer Hans Monderman.[6] In the early 2000s, Monderman introduced the "shared space"[7] model, a radical departure from conventional traffic management. His approach emphasized simplifying street design, eliminating the clutter of traffic signals, signs, and road markings. Monderman believed that an over-reliance on technology and physical barriers not only undermined people's common sense and intelligence but also eroded the collective sense of responsibility and individual connection to space. His vision was to move away from rigid control mechanisms, fostering human connectivity and communication on the streets.

However, while Monderman's approach was celebrated as a revolutionary way of rethinking traffic management and gained traction in various parts of the world, it faced challenges in certain locales. For instance, in the United Kingdom, the "shared space" design faced criticisms, particularly concerning accessibility and inclusion for people with disabilities. Research has shown that while the model aimed to enhance safety and mobility for all, it inadvertently introduced barriers for some, especially in areas where the design was not implemented with careful consideration of local contexts and needs (Lawson et al., 2022).

In a similar vein, London has recently piloted an innovative "pedestrian priority" approach (Revesz, 2022). This system defaults to a green signal for pedestrians, turning red only when vehicles approach. The underlying technology is straightforward: sensors detect moving objects from a distance. The results have been promising, enhancing street safety, saving pedestrians considerable time (about 1.3 hours), and increasing compliance with traffic signals. Moreover, the system is cost-effective, sidestepping privacy concerns inherent in more complex technologies.

Spain has leveraged similar sensor technology, albeit for a different purpose: curbing speeding on rural roads. These sensors not only alert speeding drivers (*jaydrivers*) but can also halt vehicles that flout speed limits. Drawing inspiration from these examples, Vienna has the potential to redefine its approach to traffic management. By prioritizing pedestrians and leveraging simple, effective technologies, the city can foster safer, more democratic urban spaces.

To truly democratize urban spaces and challenge the dominance of automobility, it is essential to cultivate new visions of what streets can and should be. As we transition to the concluding remarks of this article, we must consider the broader implications of these observations. How do they resonate with other similar initiatives globally? Are there socio-cultural factors that influence perceptions of jaywalking or the management of visual impairment among pedestrians in different contexts? How can these insights inform the design processes for future initiatives? And most crucially, what does this case study reveal about "mundane urban governance" and the deployment of AI-powered interventions for current and future mobility, both within cities and beyond?

Conclusion: Urban AI, A Momentum for Reimagining Urban Governance

The IPTL's emergence in Vienna offers a window into the complexities of implementing AI-driven solutions in urban settings, insights that have global resonance. As smart cities globally lean into technology for urban governance, Vienna's journey becomes a beacon (Sepehr and Felt, 2023). Cultural norms, which can vary widely, shape perceptions of practices like jaywalking. Recognizing these cultural nuances is crucial when designing AI interventions, as one-size-fits-all solutions may not always succeed. The IPTL serves as a prism, reflecting the challenges and opportunities of "mundane urban governance." As cities worldwide weave AI into their urban fabric, Vienna's experiences emphasize the importance of historical, multi-relational and inclusive design approaches that resonate with a diverse urban populace and modes of being in space. The IPTL stands as a testament to both the potential hurdles and triumphs of AI in urban governance, guiding us towards more inclusive urban technology development.

Themes of (in)visibility consistently emerged throughout the IPTL's journey. The research highlighted a gap in recognizing traffic lights as temporal regulators. This oversight became particularly evident in the neglect of the visually impaired community during the design phases. The IPTL's primary goal—eliminating the button—seemed narrow, bypassing the broader nuances of urban movement. The IPTL's debut offers a chance to challenge entrenched car-centric mobility paradigms. It also calls for a critical examination of prevailing tech-solutionist narratives, which may prioritize control over societal betterment.

The oversight of the visually impaired in the IPTL's design raises a pressing question: why were their needs overlooked? Comprehensive engagement with the multifaceted use of urban spaces is essential. Echoing Akrich's (1992) insights, this engagement should underpin design protocols, ensuring urban initiatives resonate with their socio-technical contexts. By reframing jaywalking from an individual act to a socio-technical interplay, this study underscores the pivotal role of traffic lights in the intricate ballet of urban mobility. Thus, the research advocates for embracing a broader shift to pedestrian-centric vision, enhanced by technology rather than employing an AI to fix another urban issue of the past.

Overall, the advent of AI, with its transformative potential, has ushered in a unique momentum in urban governance. This momentum is characterized by the intersection of invisibilized technology and everyday urban infrastructures. It presents an opportune moment to reopen the "black box" of technological workings, prompting urban stakeholders to critically examine the underlying mechanisms and question whom these systems truly serve. UAI, in this context, emerges not merely as a technological marvel but as a profound urban momentum—a catalyst that compels us to rethink and reconsider alternative approaches to urban life and governance.

This research underscores the importance of learning from the past, particularly in understanding how certain urban technologies create (in)visibilities in space. Such (in)visibilities, whether intentional or inadvertent, play a crucial role in shaping the design process and determining the beneficiaries of these technologies. The historical trajectory of urban technologies offers invaluable lessons, highlighting the need to prioritize inclusivity and ensure that no segment of the urban populace is marginalized or overlooked.

Traffic lights, ubiquitous in urban landscapes, subtly guide our daily rhythms. Yet, their deeper political and societal implications often remain in the background. While they primarily orchestrate traffic, they inadvertently prioritize automobility. The IPTL has ushered in a renewed examination of the intricate socio-technical dance of streets and mobility. This study delved into the IPTL initiative as both a technological artifact and a transformative agent, highlighting its role in revealing the often-hidden temporal controls within urban spaces.

Furthermore, the democratization of UAI necessitates a multi-relational and situated approach—one that is not solely reliant on technological advancements but is deeply rooted in societal values and aspirations. It calls for a collaborative endeavor, where a diverse array of stakeholders, from technologists to urban planners, and from policy-makers to everyday citizens, come together to co-create urban futures. Such collaborative efforts can ensure that AI-driven interventions are not just technologically sound but are also socially just and culturally sensitive.

In conclusion, as cities worldwide grapple with the complexities of integrating AI into their infrastructures, the lessons from Vienna and the IPTL initiative serve as a beacon. They remind us of the importance of a democratic, inclusive approach that resonates with the diverse needs and aspirations of all urban dwellers. As we stand on the cusp of a new era of AI-powered urban governance, it is imperative to approach these innovations with a critical, informed perspective, fostering open, engaging, and caring urban spaces for all.

Notes

1. Joseph Stromberg (Nov 4, 2015) explains the etymology of the word of Jaywalking on *Vox* mentioning that: "during this era, the word 'jay' meant something like 'rube' or 'hick'—a person from the sticks, who didn't know how to behave in a city. So pro-auto groups promoted use of the work 'jay walker' as someone who didn't know how to walk in a city, threatening public safety."
2. S. Goodyear, "The Invention of Jaywalking," in: *The Atlantic Cities*, 24.04.2012, http://www.theatlanticcities.com/commute/2012/04/invention-jaywalking/1837/
3. In Vienna, the surface area of the street takes up 14.5 percent of the city area and almost 25 percent of the built environment. See more here. https://www.wien.gv.at/english/administration/statistics/urban-area.html
4. See *Platz fur Wien* campaign. https://platzfuer.wien
5. Read more about the sites and where they were implemented here. https://smartcity.wien.gv.at/en/smart-traffic-lights/
6. The Guardian has a review of Monderman's traffic model and proposes it for the United Kingdom: https://www.theguardian.com/news/2008/feb/02/mainsection.obituaries
7. Read more about this project and its new development here: https://www.pps.org/article/shared-space

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Note on Contributor

Pouya Sepehr is a researcher at the Institute for Advanced Studies (IHS) Vienna, collaborating with the research group Science, Technology and Social Transformation. He also serves as a council member of the Society for Social Study of Science (4S).

ORCID

Pouya Sepehr http://orcid.org/0000-0002-4209-0759

References

M. Akrich, "The De-scription of Technical Objects," in W. Bijker and J. Law, ed., *Shaping Technology / Building Society: Studies in Sociotechnical Change* (Cambridge, MA: MIT Press, 1992) 205–224.

M. G. Ames, "Deconstructing the Algorithmic Sublime," *Big Data & Society* 5: 1 (2018) 1–4.

R. Braun, R. Randell, *Post-Automobility Futures: Technology, Power, and Imaginaries* (Maryland, MD: Rowman & Littlefield, 2022a).

R. Braun and R. Randell, "The Vermin of the Street: the Politics of Violence and the Nomos of Automobility," *Mobilities* 17: 1 (2022b) 53–68.

C. Coletta, and R. Kitchin, "Algorhythmic Governance: Regulating the 'Heartbeat' of a City Ssing the Internet of Things," *Big Data & Society* 4: 2 (2017) 1–16.

C. Coletta, T. Röhl, and S. Wagenknecht, "On Time: Temporal and Normative Orderings of Mobilities," *Mobilities* 15: 5 (2020) 635–646.

F. Cugurullo, "Urban Artificial Intelligence: From Automation to Autonomy in the Smart City," *Front. Sustain. Cities* 2: 38 (2020) 1–14.

F. Cugurullo, *Frankenstein Urbanism: Eco, Smart and Autonomous Cities, Artificial Intelligence and the End of the City* (New York: Routledge, 2021).

T. Cresswell, *On the Move: Mobility in the Modern Western World* (London: Routledge, 2006).
C. Ertler, H. Possegger, M. Opitz, and H. Bischof, "An Intent-Based Automated Traffic Light for Pedestrians." Paper presented at the 15th IEEE International Conference on Advanced Video and Signal Based Surveillance – AVSS (Auckland, 27–30 November 2018).
F. Frey, U. Leth, A. Mayerthaler, T. Brezina, "Predicted Congestions Never Occur. On the Gap Between Transport Modeling and Human Behaviour," *Transport Problems* 6 (2011) 73–86.
A. Jensen, " Mobility, Space and Power: On the Multiplicities of Seeing Mobility," *Mobilities* 6: 2 (2011) 255–271.
B. Latour, "Where Are the Missing Masses? The Sociology of a Few Mundane Artifacts," in W. Bijker and J. Law, eds, *Shaping Technology / Building Society: Studies in Sociotechnical Change* (Cambridge, MA: MIT Press, 1992) 225–259.
B. Latour, *Reassembling the Social: An Introduction to Actor-Network-Theory* (Oxford: Oxford University Press, 2005).
J. Law, *After Method: Mess in Social Science Research* (London: Routledge, 2004).
A. Lawson, I. Eskytė, M. Orchard, D. Houtzager, and E. De Vos, "Pedestrians with Disabilities and Town and City Streets: From Shared to Inclusive Space?," *The Journal of Public Space* 7: 2 (2022) 41–62. DOI: 10.32891/jps.v7i2.1603
U. Leth, F. Harald, and T. Brezina, "Innovative Approaches of Promoting Non-motorized Transport in Cities." Paper presented at the 3rd International Conference on Road and Rail Infrastructure – CETRA (Spilt Dalmatia, Croatia, May 28-30, 2014.
N. Marres, "Co-existence or Displacement: Do Street Trials of Intelligent Vehicles Test Society?" *The British Journal of Sociology* 71: 3 (2020a) 537–555.
N, Marres, "What If Nothing Happens? Street Trials of Intelligent Cars as Experiments in Participation," in S. Maasen, S. Dickel, C. Schneider, ed., *TechnoScienceSociety: Technological Reconfigurations of Science and Society*, 111–130 (Cham: Springer International Publishing, 2020b).
S. Mattern, *A City is Not a Computer: Other Urban Intelligences* (Princeton, NJ: Princeton University Press, 2021).
C. McShane, "The Origins and Globalization of Traffic Control Signals." *Journal of Urban History* 25: 3 (1999) 379–404.
S. Miyazaki, "AlgoRHYTHMS Everywhere: A Heuristic Approach to Everyday Technologies," in J. H. Hoogstad and B. Stougaard, eds. *Off Beat: Pluralizing Rhythm* (Leiden: Brill | Rodopi, 2013) 135–148.
J. Monstadt, "Urban and Infrastructural Rhythms and the Politics of Temporal Alignment," *Journal of Urban Technology* 29: 1 (2022) 69–77.
L. Mumford, *The Highway and the City* (New York: New American Library, 1964).
P. Norton, "Street Rivals: Jaywalking and the Invention of the Motor Age Street," *Technology and Culture* 48: 2 (2007) 331–359.
P. Norton, *Fighting Traffic: The Dawn of the Motor Age in the American City* (Cambridge, MA: MIT Press, 2011).
P. Norton, "Of Love Affairs and Other Stories," in S. Zavestoski and J. Agyeman, eds., *Incomplete Streets: Processes, Practices, and Possibilities* (London: Routledge, 2014) 17–35.
R. Revesz, "London is Experimenting with Traffic Lights that Put Pedestrians First," *MIT Technology Review*, June 29 2022. https://www.technologyreview.com/2022/06/29/1054565/london-traffic-lights-pedestrians/
J. Sadowski, *Too Smart: How Digital Capitalism is Extracting Data, Controlling Our Lives, and Taking Over the World*. (Cambridge, MA: MIT Press, 2020).
P. Sepehr and U. Felt, "Urban imaginaries as Tacit Governing Devices: the Case of Smart City Vienna. *ST&HV* 48:4 (2023) 1–23. https://doi.org/10.1177/01622439231178597
M. Sheller, "From Spatial Turn to Mobilities Turn," *Current Sociology* 65: 4 (2017) 623–639.
M. Sheller, *Mobility Justice: The Politics of Movement in an Age of Extremes* (London: Verso Books, 2018).
M. Sheller and J. Urry, "The City and the Car," *International Journal of Urban and Regional Research* 24: 4 (2000) 737–757.

M. Tironi and M.Valderrama, "Unpacking a Citizen Self-Tracking Device: Smartness and Idiocy in the Accumulation of Cycling Mobility Data," *Environment and Planning D: Society and Space* 36: 2 (2018) 294–312.
S. Woolgar, D. Neyland, *Mundane Governance: Ontology and Accountability* (Oxford: Oxford University Press, 2013).
V. Zeno-Zencovich, "Lessons From a Traffic Light: A Juridical Scherzo," *European Journal of Comparative Law and Governance* 3: 1 (2016) 3–23.